

CE 206: Engineering Computation Sessional

Introductory Statistics

* quantile()

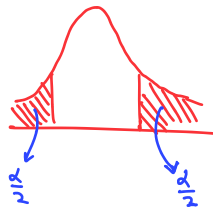
* pctlfile()

$$\bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$$

(margin)

$$[\bar{x} - \text{margin}, \bar{x} + \text{margin}]$$

* tinv()



Y axis

→ X axis

	PM 2.5	Rain	Temp	W
PM 2.5				
Rain				
Temp				
W				

Descriptive Statistics (1)

Mean: For a data set, the **arithmetic mean**, also known as "**arithmetic average**", is a **measure of central tendency** of a finite set of numbers: specifically, the sum of the values divided by the number of values.

$$\bar{x} = \frac{1}{n} \left(\sum_{i=1}^n x_i \right) = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

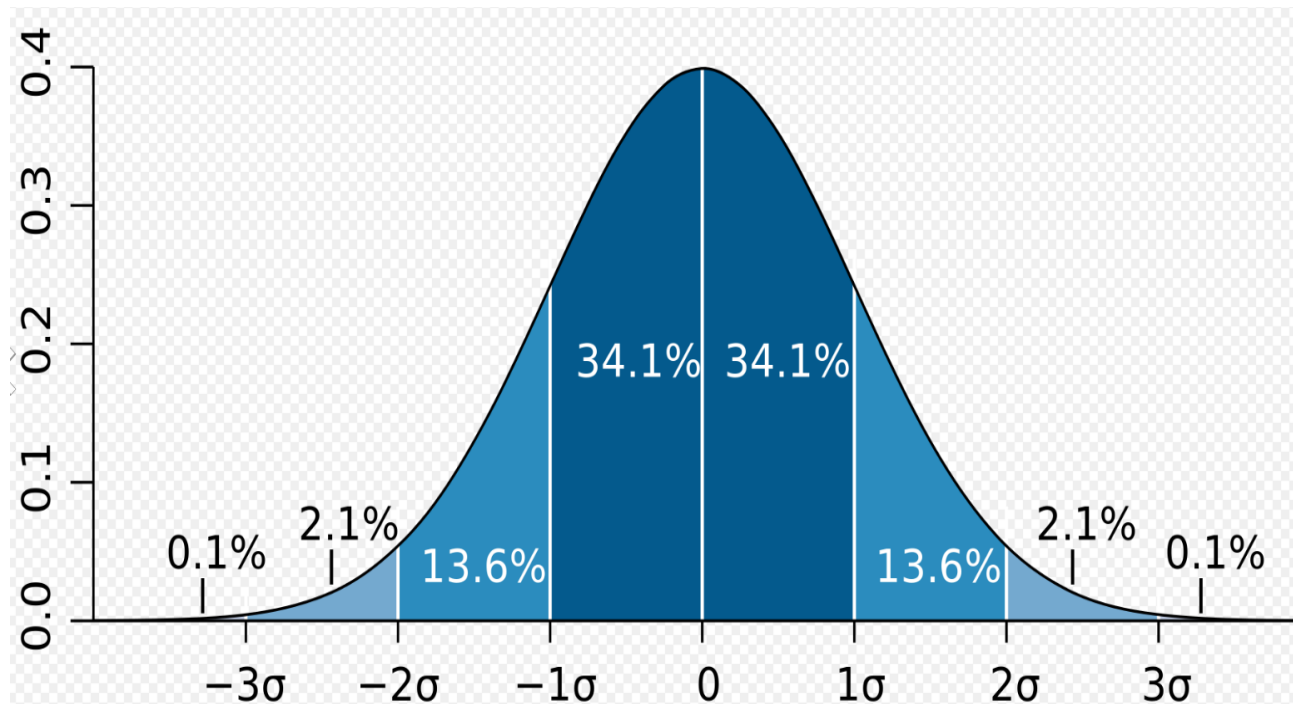
Median: the median is the **value separating** the **higher half** from the **lower half** of a **data sample**, a **population**, or a **probability distribution**. For a data set, it may be thought of as "the **middle**" value.

1, 3, 3, 6, 7, 8, 9

has the median of 6, which is the fourth value.

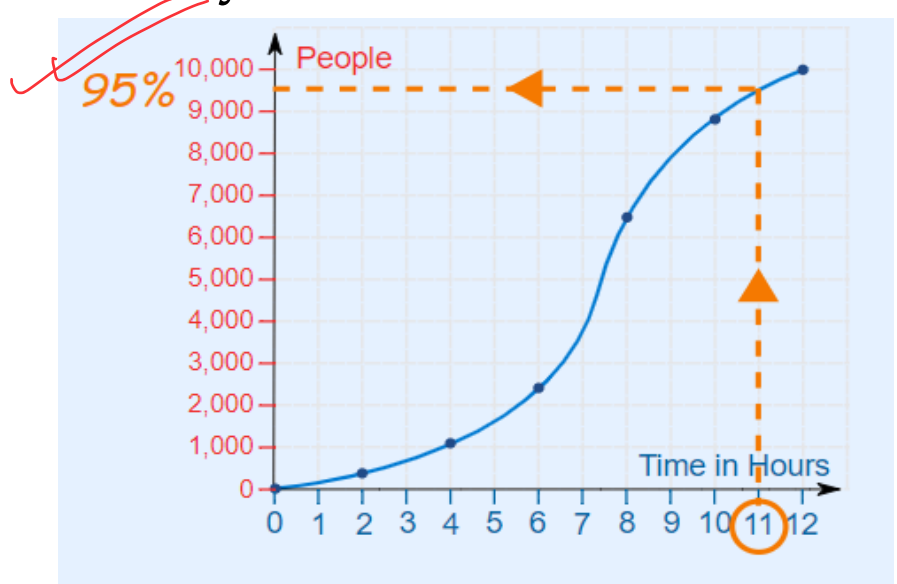
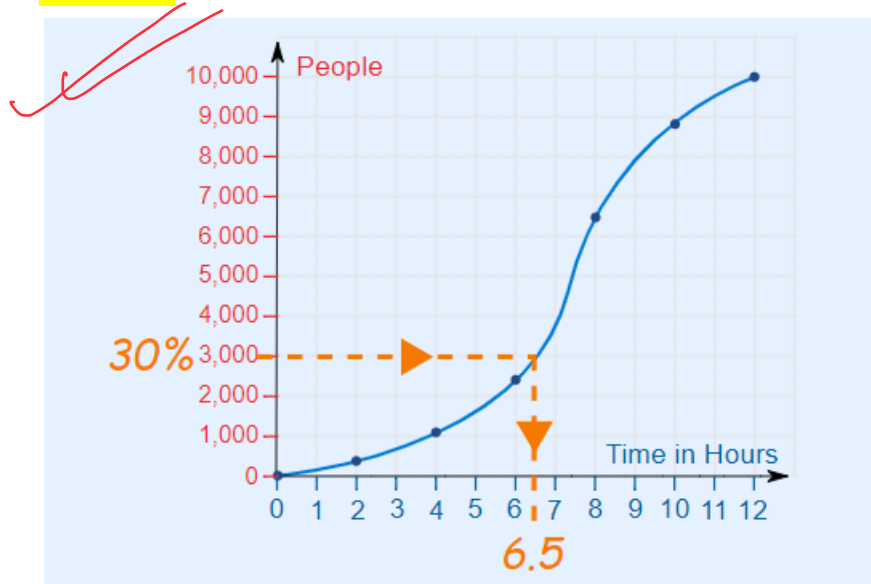
Descriptive Statistics (2)

Distribution: In probability theory and statistics, a probability **distribution** is the **mathematical function** that gives the **probabilities** of **occurrence of different possible outcomes** for an experiment, It is a **mathematical description** of a **random phenomenon** in terms of its **sample space** and the **probabilities of events**.



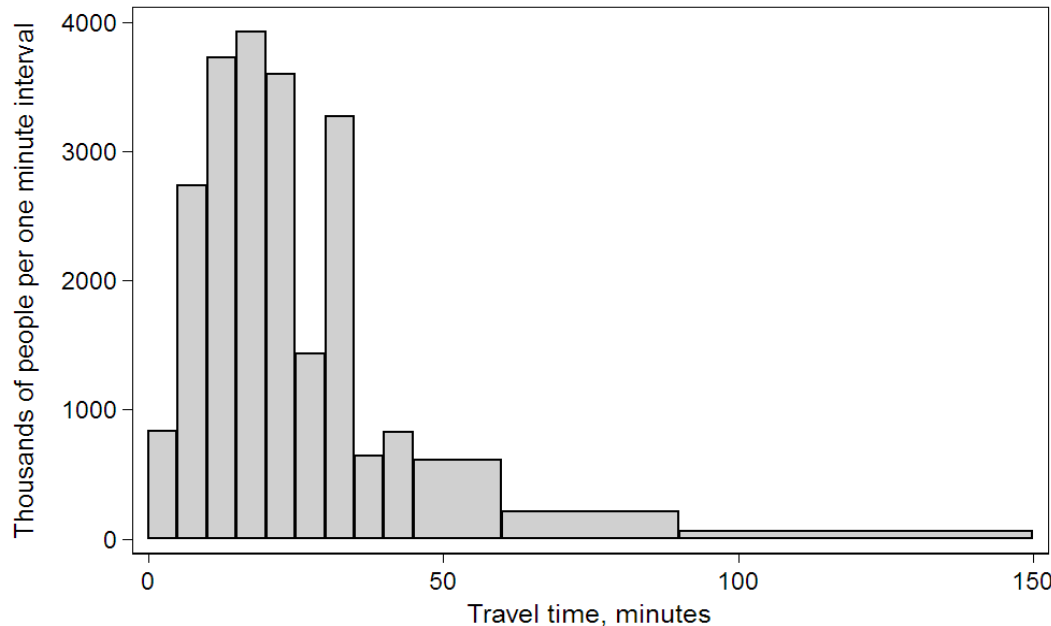
Descriptive Statistics (3)

Percentile: In statistics, a k-th percentile (percentile score or centile) is a score below which a given percentage k of scores in its frequency distribution falls (exclusive definition) or a score at or below which a given percentage falls (inclusive definition). For example, the 50th percentile (the median) is the score below which (exclusive) or at or below which (inclusive) 50% of the scores in the distribution may be found.



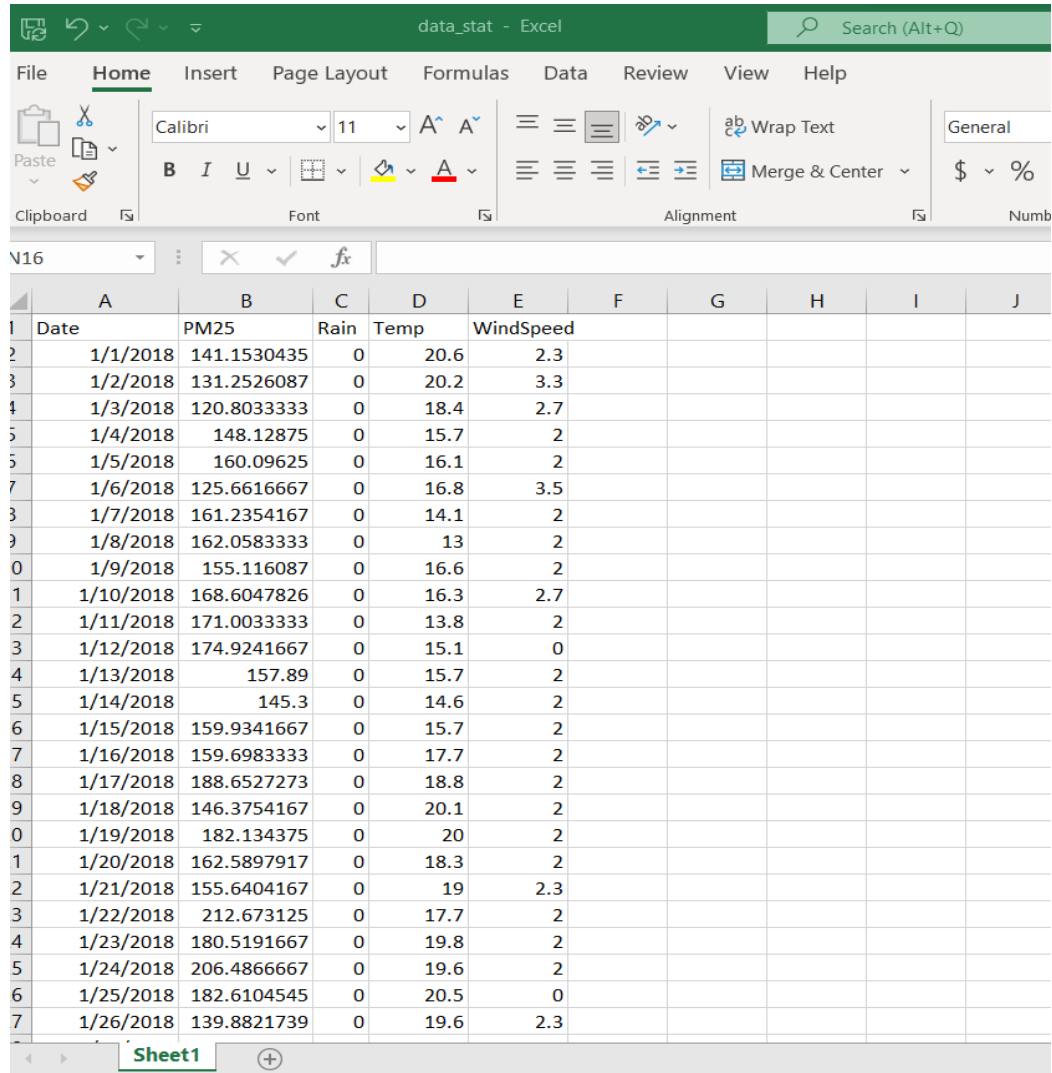
Histogram

Histogram: A histogram is an approximate representation of the distribution of numerical data. The term was first introduced by Karl Pearson. To construct a histogram, the first step is to "bin" (or "bucket") the range of values—that is, divide the entire range of values into a series of intervals—and then count how many values fall into each interval.



Example: Data Import

```
dfTable = readtable ('data_stat.xlsx')
```



The screenshot shows an Excel spreadsheet titled 'data_stat - Excel'. The ribbon is set to 'Home', and the 'Font' group is active. The spreadsheet contains a table with the following data:

	A	B	C	D	E	F	G	H	I	J
1	Date	PM25	Rain	Temp	WindSpeed					
2	1/1/2018	141.1530435	0	20.6	2.3					
3	1/2/2018	131.2526087	0	20.2	3.3					
4	1/3/2018	120.8033333	0	18.4	2.7					
5	1/4/2018	148.12875	0	15.7	2					
5	1/5/2018	160.09625	0	16.1	2					
7	1/6/2018	125.6616667	0	16.8	3.5					
3	1/7/2018	161.2354167	0	14.1	2					
9	1/8/2018	162.0583333	0	13	2					
0	1/9/2018	155.116087	0	16.6	2					
1	1/10/2018	168.6047826	0	16.3	2.7					
2	1/11/2018	171.0033333	0	13.8	2					
3	1/12/2018	174.9241667	0	15.1	0					
4	1/13/2018	157.89	0	15.7	2					
5	1/14/2018	145.3	0	14.6	2					
6	1/15/2018	159.9341667	0	15.7	2					
7	1/16/2018	159.6983333	0	17.7	2					
8	1/17/2018	188.6527273	0	18.8	2					
9	1/18/2018	146.3754167	0	20.1	2					
0	1/19/2018	182.134375	0	20	2					
1	1/20/2018	162.5897917	0	18.3	2					
2	1/21/2018	155.6404167	0	19	2.3					
3	1/22/2018	212.673125	0	17.7	2					
4	1/23/2018	180.5191667	0	19.8	2					
5	1/24/2018	206.4866667	0	19.6	2					
6	1/25/2018	182.6104545	0	20.5	0					
7	1/26/2018	139.8821739	0	19.6	2.3					

Example: Mean

```
dfTable = readtable ('data_stat.xlsx')
dfTable.PM25
ans =
141.1530
131.2526
120.8033
.....
107.1646
```

New variable can be created

```
pm25= dfTable.PM25
mean (pm25) or mean (dfTable.PM25)
```

```
ans=
100.5309
```

Example: Mean

Matrix form can be created

```
dataMat = [dfTable.PM25, dfTable.Rain,  
dfTable.Temp, dfTable.WindSpeed];
```

→ default: (columnwise means)

```
mean(dataMat)
```

```
ans=
```

```
100.5309    4.7586    26.0948    3.0003
```

If row-wise mean or other parameters are required;

`mean(A,2)` computes the mean of the elements in each row of A and returns an m-by-1 column vector.

Try

→ (no. of rows)

```
mean(dataMat,2);
```

Exercise

(0)

11

Find the following parameters for the PM_{2.5}, Rainfall, Temperature, and Windspeed of the data provided.

1. Minimum
2. Maximum
3. Standard deviation
4. 1st, 2nd, and 3rd quartile (prctile() or, quantile())
5. IQR (defined as the range between 3rd and 1st quartile)
6. 95% Confidence Interval

$t_{inv}(cl, v)$

$$\alpha = 5\%$$

$$\frac{\alpha}{2} = 2.5\%$$

$$1 - 2.5\% = 0.975$$

$$\bar{x} \pm t \cdot \frac{s}{\sqrt{n}}$$

margin

$$CI = [\bar{x}_{bar} - \text{margin}, \bar{x}_{bar} + \text{margin}]$$

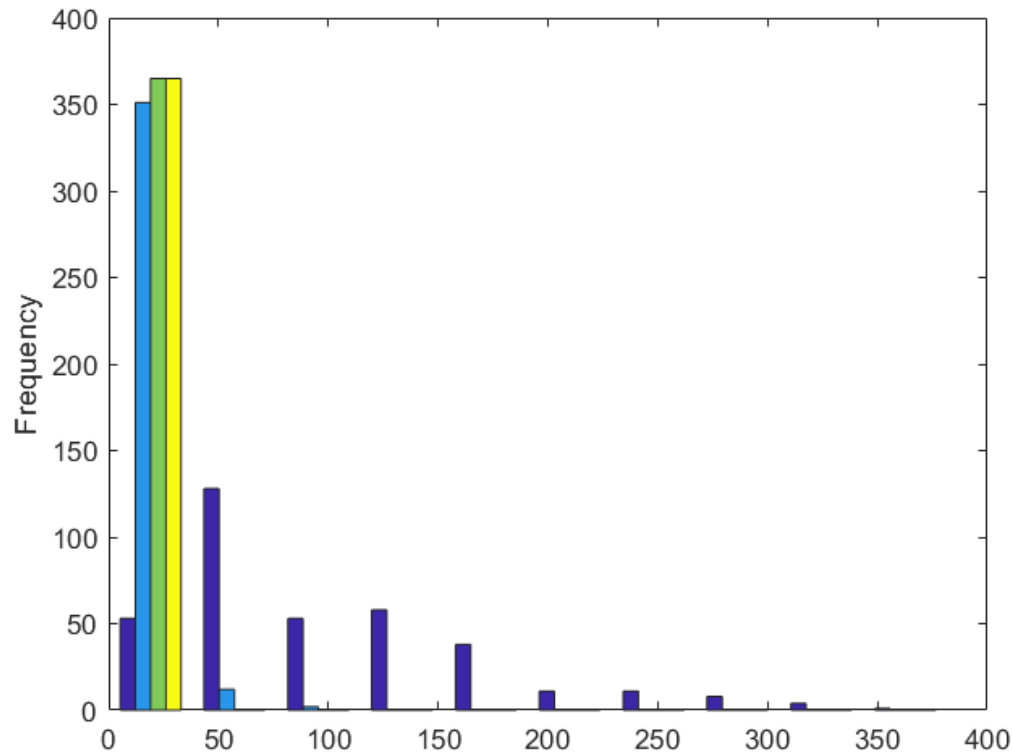
$$\begin{aligned} x &= PM_{2.5}; \quad n = \text{length}(x); \\ \bar{x}_{bar} &= \text{mean}(x); \quad s = \text{std}(x); \\ \text{margin} &= t_{inv}(0.975, n-1) * s / \sqrt{n}; \end{aligned}$$

1. Basic Syntax
The most common way to use the function is by providing a data set and a set of probabilities.

- `q = quantile(A, p)`
Returns the quantiles of data `A` for probabilities `p`. [1][2]
 - `A`: Can be a vector, matrix, or multidimensional array. [1][2]
 - `p`: A value or vector of values between 0 and 1 (e.g., 0.5 for the median, 0.25 for the first quartile).
- `q = quantile(A, q)`
Returns `q` evenly spaced quantiles [2].
 - If `q = 3`, MATLAB calculates quantiles for probabilities `1/4, 2/4, 3/4`.
 - Note: For `n` groups, it returns `n - 1` cut points (e.g., `n = 4` returns the 3 quantiles).

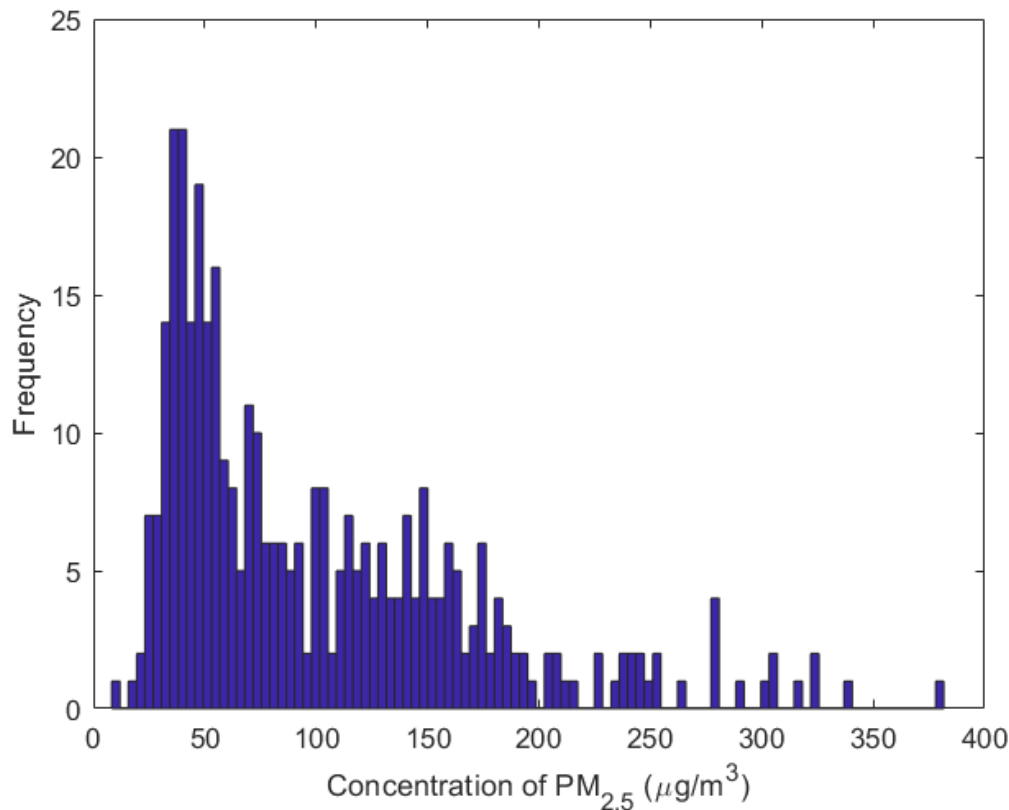
Example: Histogram

```
hist(dataMat) ; ↪ default bins = 10.  
ylab('Frequency') ; ↪ (bar intervals)
```



Example: Histogram

```
hist(pm25,100);  
ylabel('Frequency');  
xlabel('PM_{2.5} Concentration (\mu g/m^3)')
```



Plots percentile with 100 bars within maximum and minimum values. Bin number can be varied this way.

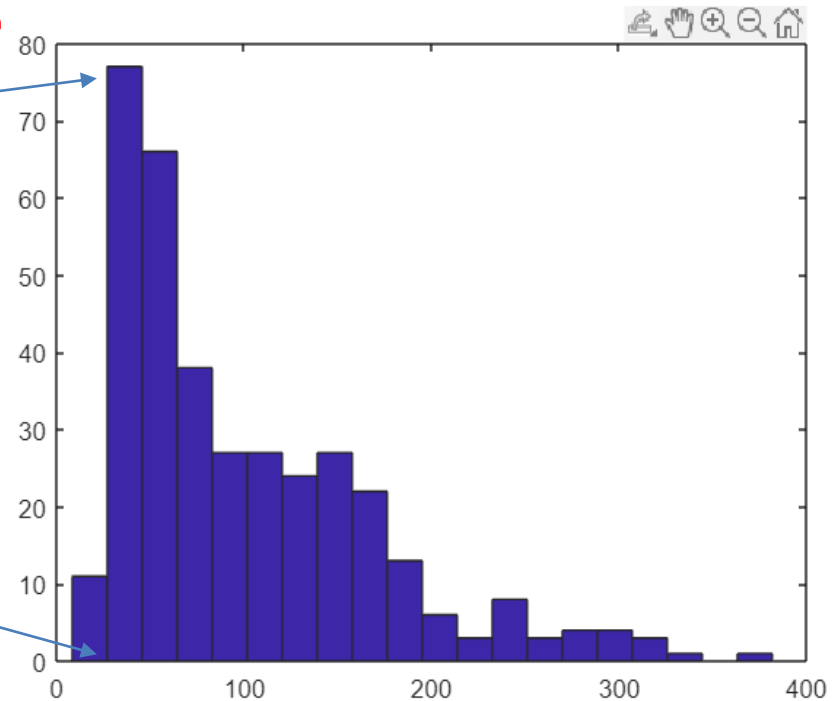
Example: Histogram

frequency of that interval *(midpoint of each bin)*
[n,x]= hist (pm25,20)
↳ eita dile only shows output ; doesn't plot histogram

Output:

n=
11 77 66

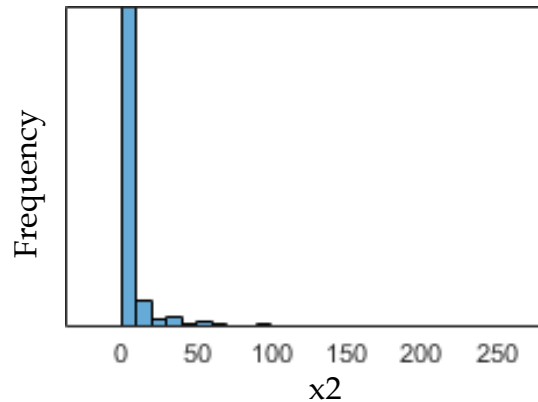
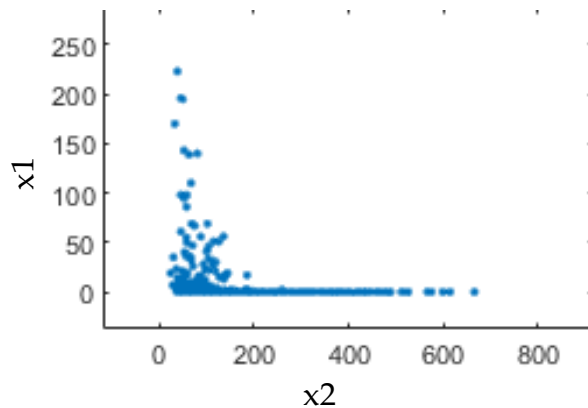
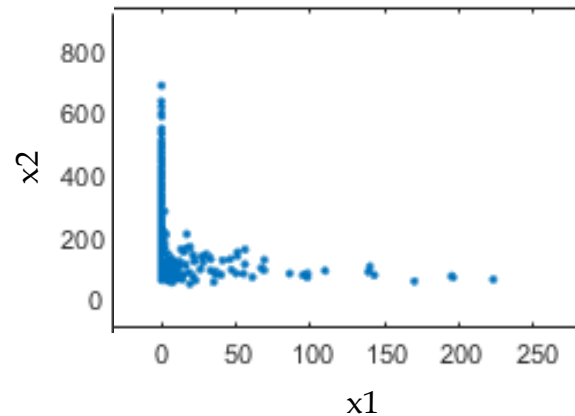
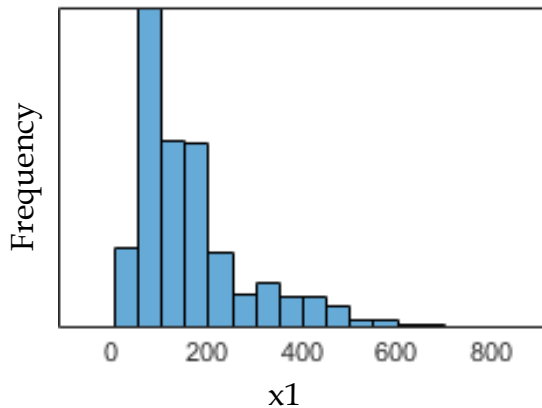
(bin centres)
x=
17.55 36.22 54.89.....



plotmatrix

The function is generally called in the following way:

```
plotmatrix(data)
```



x1	x2
139.02	17
86.56	16
96.30	22
111.18	32
119.20	30
139.02	17
86.56	16
96.30	22
⋮	⋮

Definitions

Hypothesis Testing: A statistical hypothesis test is a method of statistical inference used to decide whether the data at hand sufficiently support a particular hypothesis. Hypothesis testing allows us to make probabilistic statements about population parameters.

A **population** is the entire group that you want to draw conclusions about. A **sample** is the specific group that you will collect data from. The size of the sample is always less than the total size of the population. In research, a population doesn't always refer to people

Definitions

z test: A **z-test** is a statistical test to determine whether **two population means** are **different** when the **variances** are **known** and the **sample size is large**. A z-test is a hypothesis test in which the z-statistic follows a **normal distribution**. A **z-statistic**, or **z-score**, is a **number representing** the **result** from the **z-test**.

t test: A t-test is the most commonly applied when the test statistic would follow a **normal distribution** if the **value of a scaling term in the test statistic were known**. When the **scaling term is unknown** and is **replaced by an estimate** based on the data, the test statistics (under certain conditions) follow a Student's t distribution. The t-test can be used, for example, to **determine** if the **means of two sets of data are significantly different from each other**.  (paired t-test)

“In both cases it is assumed that distribution is normal”

Example: Hypothesis Testing (t- test)

To compare the pollution level of first weeks of the year using PM2.5 concentration

(D) |

Sample 1: January 1-7

Sample 2: January 8-14

$N=7$

t-test will be performed

$H_0 \quad \mu_1 = \mu_2$

$H_1 \quad \mu_1 \neq \mu_2$ (Two Tail Test)

Example: Hypothesis Testing (t- test)

```
x1= dfTable.PM25(1:7) ;
```

```
x2= dfTable.PM25(8:14);
```

```
[h,p,ci,stats]= ttest2(x1,x2);
```

h= Hypothesis result (paired t-test)

p= p value

ci= Confidence Interval

stats= Statistical values

h = 1 (reject the null hypothesis)

p = 0.0135

ci = -36.6987

-5.1772

stats =

struct with fields:

tstat: -2.8945

df: 12 (Dof = $n_1 + n_2 - 2$)

sd: 13.5329

(standard deviation)

$\alpha = 0.05$ (default)

```
One-Sample Z-Test
Basic Syntax
[h,p] = ztest(x, mu, sigma)

With Significance Level
[h,p] = ztest(x, mu, sigma, alpha)

With Tail Specification
[h,p] = ztest(x, mu, sigma, alpha, tail)

Inputs
• x - sample data vector
• mu - hypothesized population mean
• sigma - known population standard deviation
• alpha - significance level (default = 0.05)
• tail -
  • 'both' (default) - two-tailed
  • 'right' - right-tailed (p > mu)
  • 'left' - left-tailed (p < mu)

Outputs
• h - 1 if reject H0, 0 otherwise
• p - p-value
```

```
Two-Sample Z-Test
Used when comparing two population means with known population standard deviations.
Basic Syntax
[h,p] = ztest(x1, x2, sigma1, sigma2)

With Significance Level
[h,p] = ztest(x1, x2, sigma1, sigma2, alpha)

With Tail Specification
[h,p] = ztest(x1, x2, sigma1, sigma2, alpha, tail)

Inputs
• x1, x2 - sample data vectors
• sigma1, sigma2 - known population standard deviations
• alpha - significance level (default = 0.05)
• tail -
  • 'both' (default)
  • 'right' - p > mu1 - mu2
  • 'left' - p < mu1 - mu2

Outputs
• h - hypothesis test result
• p - p-value
```

Example: Hypothesis Testing

Test whether mean pollution level (PM2.5 concentration) is more than 100 mg/m³ for 7 days in January month.

t-test; one sample

$$H_0 \quad \mu = 100$$

$$H_a \quad \mu > 100$$

```
[h,p,ci,stats]= ttest(x1,100);
```

```
h = 1
```

```
P = 5.1456e-04
```

```
ci= 126.2636  
    156.1167
```

```
stats =
```

```
struct with fields:
```

```
tstat: 6.7523
```

```
df: 6
```

```
sd: 16.1395
```

(D) 1

z-test also available
↓
check GPT

Discuss with Simanto (Eita val myb.)

ttest (x1, 100, 'Tail', 'right')

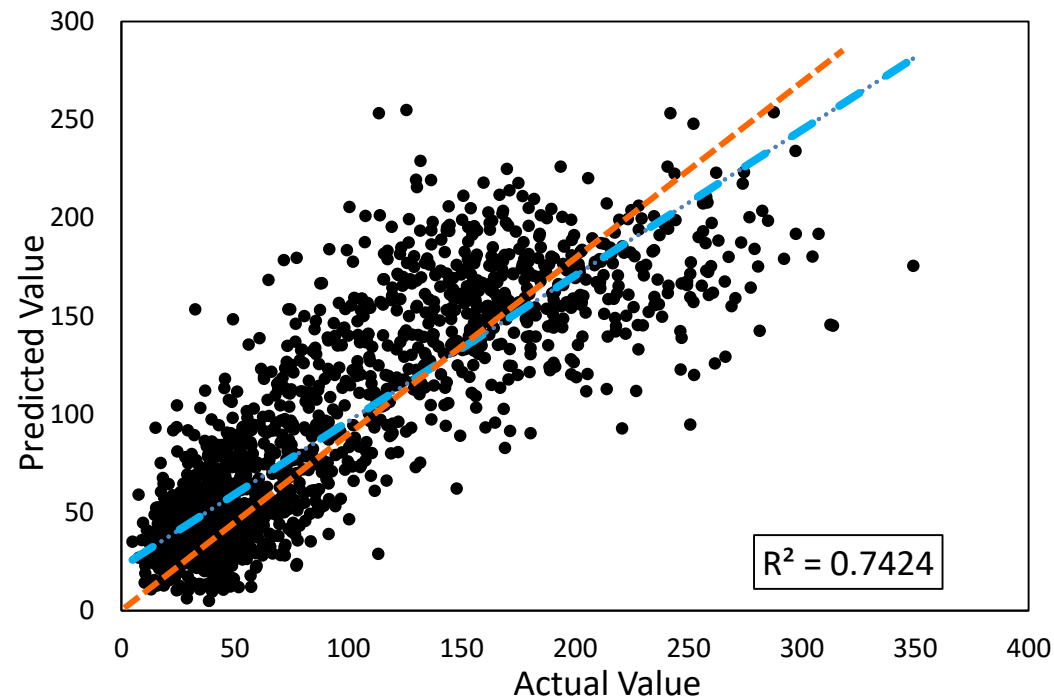
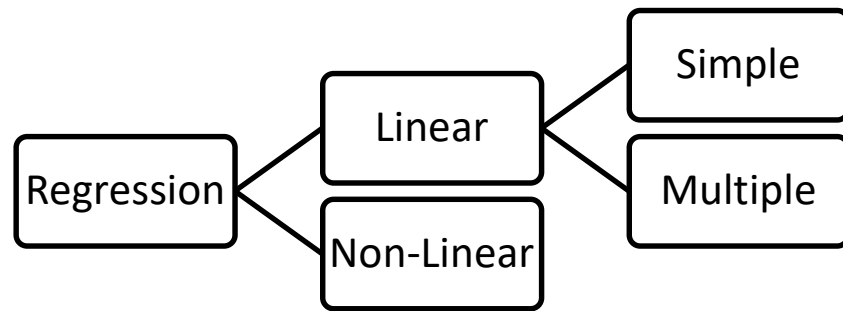
$\mu > 100 \rightarrow$ right tail

$\mu < 100 \rightarrow$ left tail

(GPT)

Regression Model

A regression model provides a function that describes the relationship between one or more independent variables and a response, dependent, or target variable.



Example: Regression Model

Conduct a linear regression using the provided data and find the association between $PM_{2.5}$ concentration and the meteorological parameters.

1. Simple Linear Regression (Association between $PM_{2.5}$ and rainfall)
2. Multiple Linear Regression (Association between $PM_{2.5}$ and all three meteorological parameters)

Data as table:

```
fitlm(dfTable, PM25~Rain)
```

Data as matrix:

```
fitlm(dataMat(:,2), dataMat(:,1))
```

Example: Simple Linear Regression

To find the correlation between PM_{2.5} concentration and Rainfall:

```
fitlm(dfTable, 'PM25~Rain')
```

Linear regression model:

PM25 ~ 1 + Rain

y intercept $\rightarrow x_0$ x

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	107.38 (x_0)	3.7754	28.443	2.0563e-94
Rain	-1.4401 (x_1)	0.29034	-4.9601	1.0845e-06

Number of observations: 365, Error degrees of freedom: 363

Root Mean Squared Error: 67.1

R-squared: 0.0635, Adjusted R-Squared: 0.0609

F-statistic vs. constant model: 24.6, p-value = 1.08e-06

Example: Simple Linear Regression

To find the correlation between PM_{2.5} concentration and Rainfall (No intercept, passing through origin):

```
fitlm(dfTable, 'PM25~Rain', 'intercept', 'false')
```

Linear regression model:

```
PM25 ~ Rain
```

$y = mx$

concept: works only in
case of data as
table format, not the
matrix one.

Estimated Coefficients:

	Estimate	SE	tStat	pValue
Rain	1.582	0.48484	3.2629	0.001207

Number of observations: 365, Error degrees of freedom: 364

Root Mean Squared Error: 120

Example: Multiple Linear Regression

To find the correlation between PM_{2.5} concentration and Rainfall, temperature, and windspeed:

```
fitlm(dfTable, 'PM25~Rain+Temp+WindSpeed')
```

Linear regression model:

```
PM25 ~ 1 + Rain + Temp + WindSpeed
```

Using matrices:

```
fitlm(dataMat(:, 2:4), dataMat(:, 1))
```

$\downarrow$ $\downarrow$
 x_1, x_2, x_3 y

Estimated Coefficients:

	Estimate	SE	tStat	pValue
(Intercept)	346.84	18.032	19.235	2.9249e-57
Rain	-1.0358	0.23509	-4.4062	1.3888e-05
Temp	-7.7203	0.71101	-10.858	5.8773e-24
WindSpeed	-13.305	2.3095	-5.7608	1.7939e-08

Number of observations: 365, Error degrees of freedom: 361

Root Mean Squared Error: 53.9

R-squared: 0.4, Adjusted R-Squared: 0.395

F-statistic vs. constant model: 80.2, p-value = 9.24e-40

Example: Multiple Linear Regression

To find the correlation between PM_{2.5} concentration and Rainfall, temperature, and windspeed (no intercept):

```
fitlm(dfTable, 'PM25~Rain+Temp+WindSpeed', 'intercept', 'false')
```

Linear regression model:

PM25 ~ Rain + Temp + WindSpeed

Estimated Coefficients:

	Estimate	SE	tStat	pValue
	_____	_____	_____	_____
Rain	-1.1995	0.33384	-3.5932	0.00037186
Temp	4.8215	0.40283	11.969	4.8174e-28
WindSpeed	-9.3537	3.2688	-2.8615	0.0044608

Number of observations: 365, Error degrees of freedom: 362

Root Mean Squared Error: 76.6

Output: Regression Model

Estimate - Coefficient estimates for each corresponding term in the model.

SE - Standard error of the coefficients.

$$\left(\frac{s}{\sqrt{n}} \right)$$

tStat - t-statistic for each coefficient to test the null hypothesis that the corresponding coefficient is zero against the alternative that it is different from zero, given the other predictors in the model. [$tStat = Estimate/SE$]

$$\rightarrow (\pi - \mu_0)$$

pValue - pValue for the t-statistic of the hypothesis test that the corresponding coefficient is equal to zero or not.
this term is not significant at the 5% significance level given the other terms in the model.